

Supplemental Code to Accompany Broda et al.'s Social Network Research Tutorial

Michael Broda, Ph.D.

August 24, 2021

Installing packages:

The following code will install the packages required for this analysis. The `repos` argument specifies a specific repository to use, and the `dependencies` argument specifies that package dependencies associated with the listed packages should also be installed.

```
install.packages(c("car", "igraph", "patchwork", "table1", "tidyverse"), repos =  
"http://cran.rstudio.com", dependencies = TRUE)
```

We need to explicitly load in 1 package:

When at all possible, we use the `package::function()` format to facilitate reproducible code in different environments. However, we need to load the `patchwork` package, which is not well suited to the `package::function()` structure. `patchwork` facilitates combining plots into combined elements.

```
library(patchwork)
```

```
## Warning: package 'patchwork' was built under R version 3.6.2
```

The pipe operator, `%>%`, from the `magrittr` package is a useful tool for piping a series of commands into one series. Here, we just load the pipe function from `magrittr` and assign it to the symbol `%>%`.

```
`%>%` <- magrittr::`%>%`
```

Import the Dataset and Use `dplyr::glimpse` to Explore:

```
langnet <- haven::read_dta("langnet.dta") %>%
  dplyr::mutate(.,
    risk_status_fac = forcats::as_factor(risk_status),
    race_eth_fac = forcats::as_factor(race_eth),
    gen_fac = forcats::as_factor(gen))
```

```
dplyr::glimpse(langnet)
```

```
## Rows: 188
## Columns: 14
## $ student_id      <chr> "10108", "10117", "10119", "10102", "10114", "10107", ...
## $ teachid         <dbl+lbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
## $ bf1             <chr> "10120", "10101", "NA", "10117", "10106", "10116", "NA..."
## $ bf2             <chr> "NA", "10102", "NA", "10108", "10103", "10104", "NA", ...
## $ bf3             <chr> "NA", "10119", "NA", "10120", "10119", "10105", "NA", ...
## $ risk_status      <dbl+lbl> 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, ...
## $ celf_score       <dbl> 14, 10, 12, 12, 2, 11, 8, 12, 12, 15, 4, 12, 7, 10, 15...
## $ race_eth         <dbl+lbl> 1, 1, 1, 1, 5, 1, 1, 1, 1, 1, 1, 1, 1, 5, 1, 1, ...
## $ gen              <dbl+lbl> 2, 1, 2, 1, 2, 1, 2, 2, 2, 1, 1, 2, 1, 2, 1, 2, ...
## $ csize            <dbl> 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22...
## $ age_yrs          <dbl> 6.592745, 6.516085, 6.425736, 7.071868, 6.710472, 6.34...
## $ risk_status_fac  <fct> 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, ...
## $ race_eth_fac     <fct> WHITE, WHITE, WHITE, WHITE, OTHER, WHITE, WHITE, WHITE...
## $ gen_fac          <fct> MALE, FEMALE, MALE, FEMALE, MALE, FEMALE, MALE, MALE, ...
```

Create a Student-Level Descriptive Table, by Lang. Risk Status

```
table1::table1(~ age_yrs + race_eth_fac + gen_fac + celf_score | risk_status_fac, data = langnet)
```

```
## Get nicer `table1` .docx output by simply installing the `flectable` package
```

		0	1	Overall
		(N=112)	(N=76)	(N=188)
## 1				
## 2	age_yrs			
## 3	Mean (SD)	6.77 (0.318)	6.88 (0.340)	6.81 (0.330)

```
## 4   Median [Min, Max] 6.73 [6.25, 7.65] 6.91 [6.26, 7.49] 6.85 [6.25, 7.65]
## 5       race_eth_fac
## 6           WHITE           80 (71.4%)           40 (52.6%)           120 (63.8%)
## 7   AFRICAN AMERICAN           16 (14.3%)           12 (15.8%)           28 (14.9%)
## 8           ASIAN             2 (1.8%)             2 (2.6%)             4 (2.1%)
## 9           HISPANIC           2 (1.8%)           11 (14.5%)           13 (6.9%)
## 10          OTHER            10 (8.9%)            8 (10.5%)           18 (9.6%)
## 11          Missing           2 (1.8%)            3 (3.9%)            5 (2.7%)
## 12          gen_fac
## 13          FEMALE           53 (47.3%)           32 (42.1%)           85 (45.2%)
## 14          MALE            59 (52.7%)           44 (57.9%)          103 (54.8%)
## 15          celf_score
## 16          Mean (SD)        11.2 (2.29)          7.01 (2.09)          9.53 (3.03)
## 17   Median [Min, Max] 11.0 [5.00, 18.0] 7.00 [2.00, 13.0] 9.00 [2.00, 18.0]
```

Step 2: Visualizing Classroom Networks

We can select classrooms from our edgelist and create some graphs. First, we break our edgelist up into 21 groups, one for each classroom.

Classroom 0101

```
friends0101 <- dplyr::filter(langnet, stringr::str_detect(student_id, "^101"))
nodes0101 <- dplyr::select(friends0101, student_id, risk_status)
ties0101 <- dplyr::select(friends0101, nom = student_id, bf1, bf2, bf3) %>%
  tidyr::pivot_longer(-nom, values_to = "nom2") %>%
  dplyr::filter(., nom2 != "NA") %>%
  dplyr::select(., -name)

g0101 <- igraph::graph_from_data_frame(ties0101, directed = TRUE, vertices = nodes0101)

## Warning in if (class(newval) == "factor") {: the condition has length > 1 and
## only the first element will be used

g0101
```

```

## IGRAPH 01b0b9e DN-- 22 49 --
## + attr: name (v/c), risk_status (v/n)
## + edges from 01b0b9e (vertex names):
## [1] 10108->10120 10117->10101 10117->10102 10117->10119 10102->10117
## [6] 10102->10108 10102->10120 10114->10106 10114->10103 10114->10119
## [11] 10107->10116 10107->10104 10107->10105 10105->10118 10105->10121
## [16] 10105->10107 10110->10119 10110->10116 10110->10118 10104->10107
## [21] 10104->10105 10104->10106 10113->10109 10113->10107 10113->10120
## [26] 10103->10108 10103->10112 10103->10110 10106->10108 10106->10109
## [31] 10106->10121 10115->10118 10115->10112 10115->10105 10101->10117
## [36] 10101->10107 10101->10104 10118->10105 10118->10117 10118->10116
## + ... omitted several edges

# Subset vertices and edges
igraph::V(g0101)

## + 22/22 vertices, named, from 01b0b9e:
## [1] 10108 10117 10119 10102 10114 10107 10109 10105 10110 10104 10113 10103
## [13] 10106 10115 10101 10118 10120 10116 10111 10112 10122 10121

igraph::E(g0101)

## + 49/49 edges from 01b0b9e (vertex names):
## [1] 10108->10120 10117->10101 10117->10102 10117->10119 10102->10117
## [6] 10102->10108 10102->10120 10114->10106 10114->10103 10114->10119
## [11] 10107->10116 10107->10104 10107->10105 10105->10118 10105->10121
## [16] 10105->10107 10110->10119 10110->10116 10110->10118 10104->10107
## [21] 10104->10105 10104->10106 10113->10109 10113->10107 10113->10120
## [26] 10103->10108 10103->10112 10103->10110 10106->10108 10106->10109
## [31] 10106->10121 10115->10118 10115->10112 10115->10105 10101->10117
## [36] 10101->10107 10101->10104 10118->10105 10118->10117 10118->10116
## [41] 10120->10108 10120->10107 10120->10104 10116->10109 10111->10108
## [46] 10111->10103 10111->10116 10112->10122 10112->10110

# Count number of edges
igraph::ecount(g0101)

```

```
## [1] 49

# Count number of vertices
igraph::vcount(g0101)

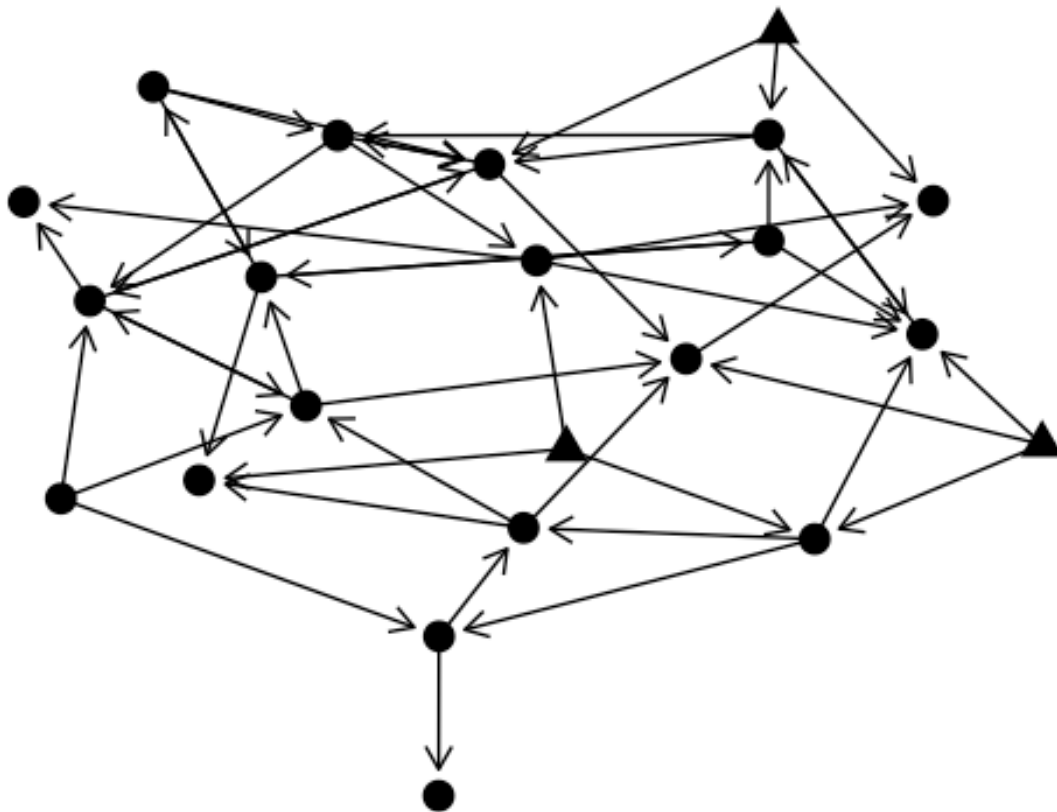
## [1] 22

#Assign risk status as a vertex attribute
igraph::V(g0101)$risk <- as.character(nodes0101$risk_status)

class101 <- ggraph::ggraph(g0101, layout = "with_kk") +
  ggraph::geom_edge_link(arrow = ggplot2::arrow(length = ggplot2::unit(3, 'mm'), type = "open", ends =
"last"),
                        end_cap = ggraph::circle(3, 'mm')) +
  ggraph::geom_node_point(ggplot2::aes(shape = risk, size = 2)) + ggplot2::theme_void() +
ggplot2::theme(legend.position = "none") + ggplot2::ggtitle("Classroom 101")

class101
```

Classroom 101



Classroom 0103

```
friends0103 <- dplyr::filter(langnet, stringr::str_detect(student_id, "^103"))  
  
nodes0103 <- dplyr::select(friends0103, student_id, risk_status)  
  
ties0103 <- dplyr::select(friends0103, nom = student_id, bf1, bf2, bf3) %>%  
  tidyr::pivot_longer(-nom, values_to = "nom2") %>%  
  dplyr::filter(., nom2 != "NA") %>%
```

```

dplyr::select(., -name)

g0103 <- igraph::graph_from_data_frame(ties0103, directed = TRUE, vertices = nodes0103)

## Warning in if (class(newval) == "factor") {: the condition has length > 1 and
## only the first element will be used

g0103

## IGRAPH 12788c4 DN-- 23 60 --
## + attr: name (v/c), risk_status (v/n)
## + edges from 12788c4 (vertex names):
## [1] 10313->10322 10313->10316 10313->10320 10315->10305 10315->10321
## [6] 10315->10318 10318->10322 10318->10323 10318->10301 10322->10323
## [11] 10322->10318 10319->10301 10319->10304 10319->10305 10323->10320
## [16] 10305->10301 10305->10320 10305->10315 10308->10305 10308->10314
## [21] 10309->10320 10309->10301 10309->10323 10301->10321 10301->10310
## [26] 10301->10318 10302->10310 10302->10321 10302->10304 10316->10302
## [31] 10316->10307 10316->10311 10317->10308 10317->10322 10321->10310
## [36] 10321->10302 10321->10303 10303->10301 10303->10321 10303->10310
## + ... omitted several edges

# Subset vertices and edges
igraph::V(g0103)

## + 23/23 vertices, named, from 12788c4:
## [1] 10313 10315 10318 10322 10319 10323 10305 10308 10309 10301 10302 10316
## [13] 10317 10321 10303 10311 10312 10320 10304 10314 10310 10307 10306

igraph::E(g0103)

## + 60/60 edges from 12788c4 (vertex names):
## [1] 10313->10322 10313->10316 10313->10320 10315->10305 10315->10321
## [6] 10315->10318 10318->10322 10318->10323 10318->10301 10322->10323
## [11] 10322->10318 10319->10301 10319->10304 10319->10305 10323->10320
## [16] 10305->10301 10305->10320 10305->10315 10308->10305 10308->10314
## [21] 10309->10320 10309->10301 10309->10323 10301->10321 10301->10310

```

```

## [26] 10301->10318 10302->10310 10302->10321 10302->10304 10316->10302
## [31] 10316->10307 10316->10311 10317->10308 10317->10322 10321->10310
## [36] 10321->10302 10321->10303 10303->10301 10303->10321 10303->10310
## [41] 10311->10307 10312->10315 10312->10305 10312->10311 10320->10301
## [46] 10320->10305 10320->10304 10304->10319 10304->10308 10304->10305
## + ... omitted several edges

# Count number of edges
igraph::ecount(g0103)

## [1] 60

# Count number of vertices
igraph::vcount(g0103)

## [1] 23

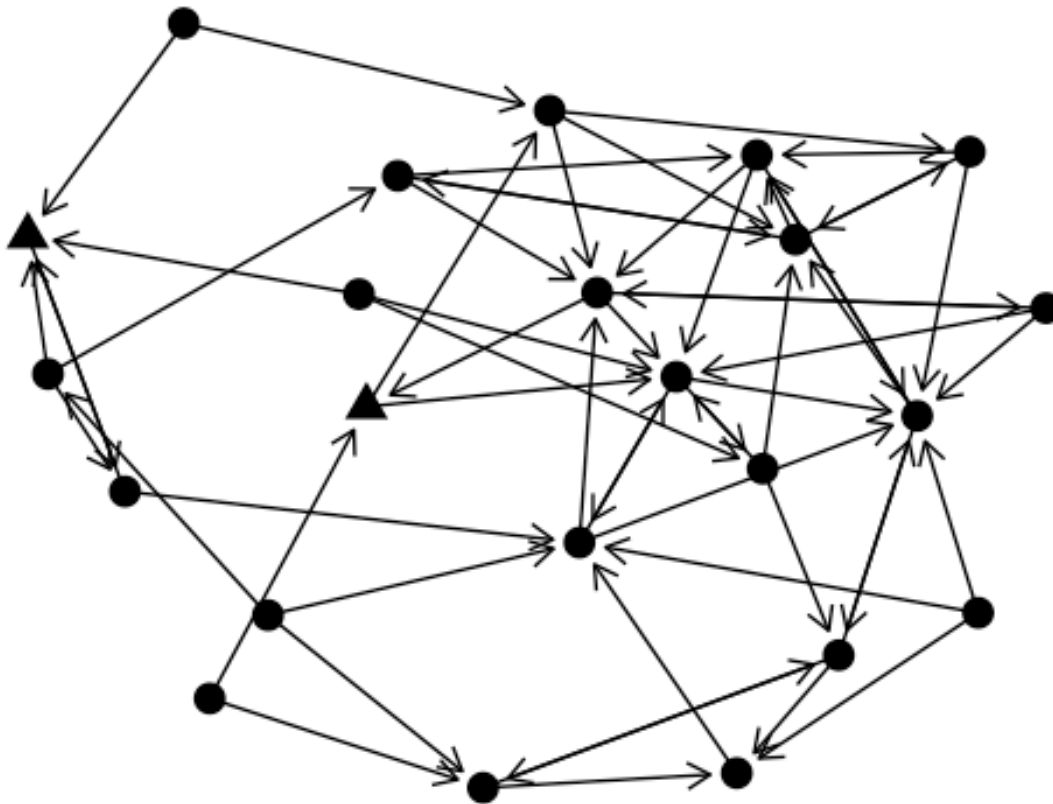
#Assign risk status as a vertex attribute
igraph::V(g0103)$risk <- as.character(nodes0103$risk_status)

class103 <- ggraph::ggraph(g0103, layout = "with_kk") +
  ggraph::geom_edge_link(arrow = ggplot2::arrow(length = ggplot2::unit(3, 'mm'), type = "open", ends =
"last"),
                        end_cap = ggraph::circle(3, 'mm')) +
  ggraph::geom_node_point(ggplot2::aes(shape = risk, size = 2)) + ggplot2::theme_void() +
ggplot2::theme(legend.position = "none") + ggplot2::ggtitle("Classroom 103")

class103

```

Classroom 103



Classroom 0105

```
friends0105 <- dplyr::filter(langnet, stringr::str_detect(student_id, "^105"))

nodes0105 <- dplyr::select(friends0105, student_id, risk_status)

ties0105 <- dplyr::select(friends0105, nom = student_id, bf1, bf2, bf3) %>%
  tidyr::pivot_longer(-nom, values_to = "nom2") %>%
  dplyr::filter(., nom2 != "NA") %>%
```

```

dplyr::select(., -name)

g0105 <- igraph::graph_from_data_frame(ties0105, directed = TRUE, vertices = nodes0105)

## Warning in if (class(newval) == "factor") {: the condition has length > 1 and
## only the first element will be used

g0105

## IGRAPH beb9e84 DN-- 16 42 --
## + attr: name (v/c), risk_status (v/n)
## + edges from beb9e84 (vertex names):
## [1] 10510->10507 10510->10516 10510->10505 10516->10505 10516->10510
## [6] 10505->10510 10505->10508 10507->10503 10507->10516 10507->10505
## [11] 10514->10505 10514->10513 10514->10512 10508->10510 10508->10512
## [16] 10515->10513 10515->10503 10513->10502 10513->10506 10513->10512
## [21] 10504->10513 10504->10514 10504->10507 10502->10513 10502->10501
## [26] 10502->10503 10501->10502 10501->10504 10501->10513 10503->10513
## [31] 10503->10507 10503->10502 10512->10513 10512->10516 10509->10504
## [36] 10509->10505 10509->10508 10506->10507 10506->10502 10506->10501
## + ... omitted several edges

# Subset vertices and edges
igraph::V(g0105)

## + 16/16 vertices, named, from beb9e84:
## [1] 10510 10516 10505 10507 10514 10508 10515 10513 10504 10502 10501 10503
## [13] 10512 10509 10506 10511

igraph::E(g0105)

## + 42/42 edges from beb9e84 (vertex names):
## [1] 10510->10507 10510->10516 10510->10505 10516->10505 10516->10510
## [6] 10505->10510 10505->10508 10507->10503 10507->10516 10507->10505
## [11] 10514->10505 10514->10513 10514->10512 10508->10510 10508->10512
## [16] 10515->10513 10515->10503 10513->10502 10513->10506 10513->10512
## [21] 10504->10513 10504->10514 10504->10507 10502->10513 10502->10501

```

```

## [26] 10502->10503 10501->10502 10501->10504 10501->10513 10503->10513
## [31] 10503->10507 10503->10502 10512->10513 10512->10516 10509->10504
## [36] 10509->10505 10509->10508 10506->10507 10506->10502 10506->10501
## [41] 10511->10512 10511->10514

# Count number of edges
igraph::ecount(g0105)

## [1] 42

# Count number of vertices
igraph::vcount(g0105)

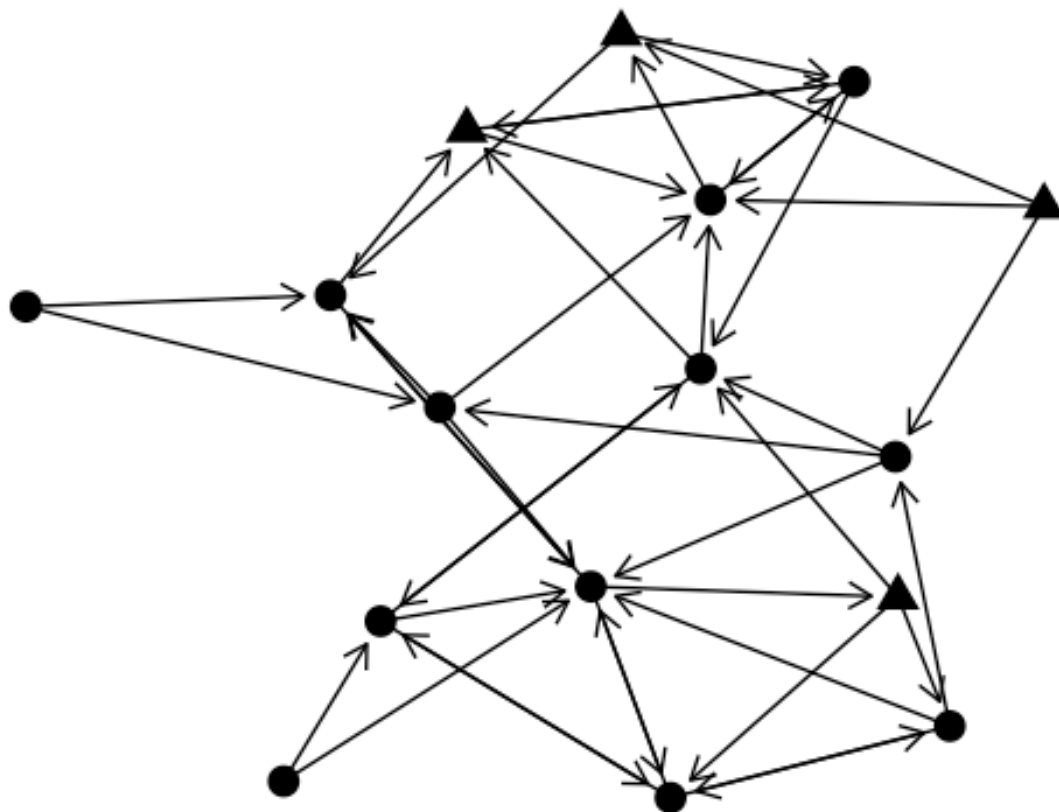
## [1] 16

#Assign risk status as a vertex attribute
igraph::V(g0105)$risk <- as.character(nodes0105$risk_status)

ggraph::ggraph(g0105, layout = "with_kk") +
  ggraph::geom_edge_link(arrow = ggplot2::arrow(length = ggplot2::unit(3, 'mm'), type = "open", ends =
"last"),
                        end_cap = ggraph::circle(3, 'mm')) +
  ggraph::geom_node_point(ggplot2::aes(shape = risk, size = 2)) + ggplot2::theme_void() +
ggplot2::theme(legend.position = "none") + ggplot2::ggtitle("Classroom 105")

```

Classroom 105



Classroom 0106

```
friends0106 <- dplyr::filter(langnet, stringr::str_detect(student_id, "^106"))  
  
nodes0106 <- dplyr::select(friends0106, student_id, risk_status)  
  
ties0106 <- dplyr::select(friends0106, nom = student_id, bf1, bf2, bf3) %>%  
  tidyr::pivot_longer(-nom, values_to = "nom2") %>%  
  dplyr::filter(., nom2 != "NA") %>%
```

```

dplyr::select(., -name)

g0106 <- igraph::graph_from_data_frame(ties0106, directed = TRUE, vertices = nodes0106)

## Warning in if (class(newval) == "factor") {: the condition has length > 1 and
## only the first element will be used

g0106

## IGRAPH ec951ef DN-- 23 62 --
## + attr: name (v/c), risk_status (v/n)
## + edges from ec951ef (vertex names):
## [1] 10612->10610 10612->10617 10612->10609 10603->10610 10603->10614
## [6] 10603->10607 10610->10612 10610->10611 10610->10604 10602->10612
## [11] 10615->10616 10615->10617 10615->10618 10611->10607 10611->10617
## [16] 10611->10610 10609->10605 10609->10612 10609->10607 10617->10607
## [21] 10617->10608 10617->10611 10605->10612 10605->10609 10605->10617
## [26] 10604->10618 10604->10621 10604->10607 10608->10613 10608->10621
## [31] 10608->10606 10606->10613 10606->10601 10606->10623 10613->10617
## [36] 10613->10609 10607->10611 10607->10612 10607->10617 10618->10611
## + ... omitted several edges

# Subset vertices and edges
igraph::V(g0106)

## + 23/23 vertices, named, from ec951ef:
## [1] 10612 10603 10610 10602 10615 10611 10609 10617 10605 10604 10608 10606
## [13] 10613 10607 10618 10620 10614 10619 10616 10621 10622 10623 10601

igraph::E(g0106)

## + 62/62 edges from ec951ef (vertex names):
## [1] 10612->10610 10612->10617 10612->10609 10603->10610 10603->10614
## [6] 10603->10607 10610->10612 10610->10611 10610->10604 10602->10612
## [11] 10615->10616 10615->10617 10615->10618 10611->10607 10611->10617
## [16] 10611->10610 10609->10605 10609->10612 10609->10607 10617->10607
## [21] 10617->10608 10617->10611 10605->10612 10605->10609 10605->10617

```

```

## [26] 10604->10618 10604->10621 10604->10607 10608->10613 10608->10621
## [31] 10608->10606 10606->10613 10606->10601 10606->10623 10613->10617
## [36] 10613->10609 10607->10611 10607->10612 10607->10617 10618->10611
## [41] 10618->10604 10618->10607 10620->10601 10620->10607 10620->10613
## [46] 10614->10603 10614->10601 10614->10602 10619->10620 10619->10618
## + ... omitted several edges

# Count number of edges
igraph::ecount(g0106)

## [1] 62

# Count number of vertices
igraph::vcount(g0106)

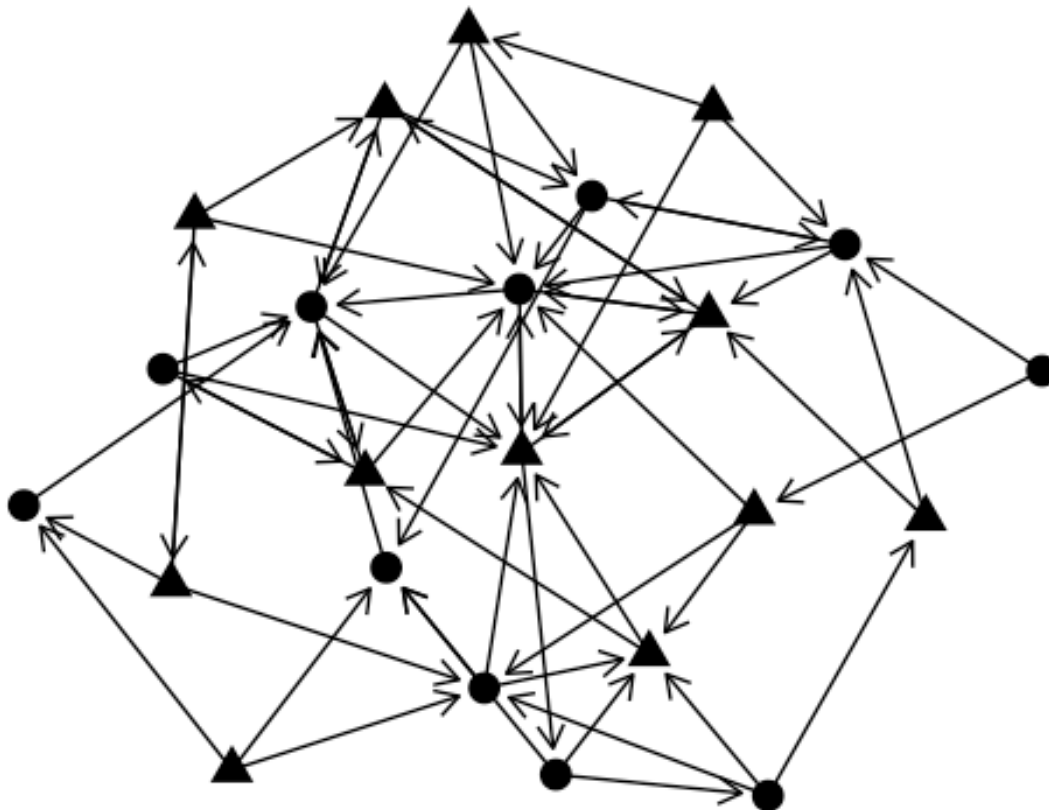
## [1] 23

#Assign risk status as a vertex attribute
igraph::V(g0106)$risk <- as.character(nodes0106$risk_status)

ggraph::ggraph(g0106, layout = "with_kk") +
  ggraph::geom_edge_link(arrow = ggplot2::arrow(length = ggplot2::unit(3, 'mm'), type = "open", ends =
"last"),
                        end_cap = ggraph::circle(3, 'mm')) +
  ggraph::geom_node_point(ggplot2::aes(shape = risk, size = 2)) + ggplot2::theme_void() +
ggplot2::theme(legend.position = "none") + ggplot2::ggtitle("Classroom 106")

```

Classroom 106



Classroom 0201

```
friends0201 <- dplyr::filter(langnet, stringr::str_detect(student_id, "^201"))  
  
nodes0201 <- dplyr::select(friends0201, student_id, risk_status)  
  
ties0201 <- dplyr::select(friends0201, nom = student_id, bf1, bf2, bf3) %>%  
  tidyr::pivot_longer(-nom, values_to = "nom2") %>%  
  dplyr::filter(., nom2 != "NA") %>%
```

```

dplyr::select(., -name)

g0201 <- igraph::graph_from_data_frame(ties0201, directed = TRUE, vertices = nodes0201)

## Warning in if (class(newval) == "factor") {: the condition has length > 1 and
## only the first element will be used

g0201

## IGRAPH a7ebb28 DN-- 22 42 --
## + attr: name (v/c), risk_status (v/n)
## + edges from a7ebb28 (vertex names):
## [1] 20121->20101 20121->20108 20120->20119 20120->20111 20110->20116
## [6] 20110->20102 20101->20111 20101->20122 20101->20121 20118->20110
## [11] 20118->20119 20113->20112 20113->20106 20114->20109 20107->20116
## [16] 20107->20122 20107->20111 20106->20109 20116->20101 20116->20112
## [21] 20116->20108 20111->20122 20111->20104 20111->20102 20109->20104
## [26] 20109->20118 20109->20112 20108->20111 20108->20102 20102->20111
## [31] 20102->20108 20102->20109 20115->20111 20115->20104 20115->20101
## [36] 20104->20115 20104->20119 20104->20116 20105->20104 20122->20111
## + ... omitted several edges

# Subset vertices and edges
igraph::V(g0201)

## + 22/22 vertices, named, from a7ebb28:
## [1] 20121 20120 20110 20101 20118 20113 20114 20107 20103 20106 20116 20111
## [13] 20109 20119 20108 20117 20102 20112 20115 20104 20105 20122

igraph::E(g0201)

## + 42/42 edges from a7ebb28 (vertex names):
## [1] 20121->20101 20121->20108 20120->20119 20120->20111 20110->20116
## [6] 20110->20102 20101->20111 20101->20122 20101->20121 20118->20110
## [11] 20118->20119 20113->20112 20113->20106 20114->20109 20107->20116
## [16] 20107->20122 20107->20111 20106->20109 20116->20101 20116->20112
## [21] 20116->20108 20111->20122 20111->20104 20111->20102 20109->20104

```

```

## [26] 20109->20118 20109->20112 20108->20111 20108->20102 20102->20111
## [31] 20102->20108 20102->20109 20115->20111 20115->20104 20115->20101
## [36] 20104->20115 20104->20119 20104->20116 20105->20104 20122->20111
## [41] 20122->20107 20122->20104

# Count number of edges
igraph::ecount(g0201)

## [1] 42

# Count number of vertices
igraph::vcount(g0201)

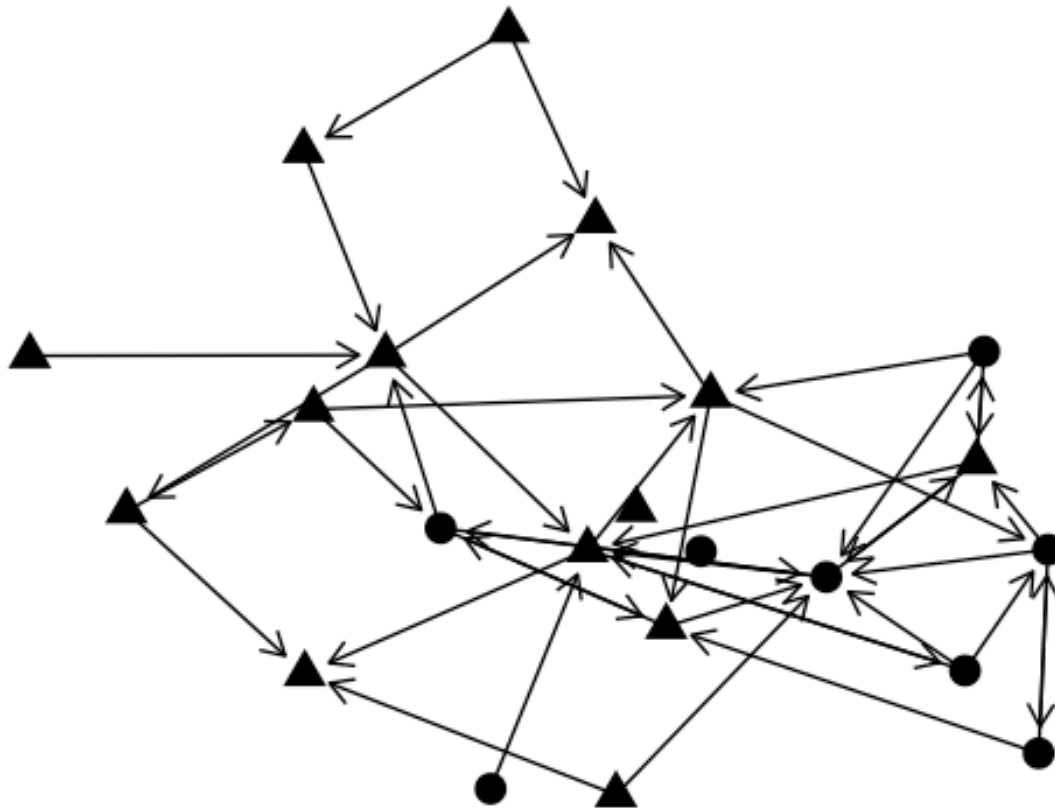
## [1] 22

#Assign risk status as a vertex attribute
igraph::V(g0201)$risk <- as.character(nodes0201$risk_status)

class201 <- ggraph::ggraph(g0201, layout = "with_kk") +
  ggraph::geom_edge_link(arrow = ggplot2::arrow(length = ggplot2::unit(3, 'mm'), type = "open", ends =
"last"),
                        end_cap = ggraph::circle(3, 'mm')) +
  ggraph::geom_node_point(ggplot2::aes(shape = risk, size = 2)) + ggplot2::theme_void() +
ggplot2::theme(legend.position = "none") + ggplot2::ggtitle("Classroom 201")

class201

```



Classroom 0203

```
friends0203 <- dplyr::filter(langnet, stringr::str_detect(student_id, "^203"))
```

```
nodes0203 <- dplyr::select(friends0203, student_id, risk_status)
```

```
ties0203 <- dplyr::select(friends0203, nom = student_id, bf1, bf2, bf3) %>%
  tidyr::pivot_longer(-nom, values_to = "nom2") %>%
  dplyr::filter(., nom2 != "NA") %>%
```

```

dplyr::select(., -name)

g0203 <- igraph::graph_from_data_frame(ties0203, directed = TRUE, vertices = nodes0203)

## Warning in if (class(newval) == "factor") {: the condition has length > 1 and
## only the first element will be used

g0203

## IGRAPH c9a1236 DN-- 18 33 --
## + attr: name (v/c), risk_status (v/n)
## + edges from c9a1236 (vertex names):
## [1] 20304->20318 20304->20317 20303->20301 20305->20313 20305->20301
## [6] 20316->20313 20316->20302 20316->20305 20320->20316 20320->20302
## [11] 20308->20304 20308->20317 20302->20306 20302->20305 20302->20314
## [16] 20312->20301 20312->20317 20306->20302 20318->20317 20318->20304
## [21] 20318->20319 20307->20304 20307->20319 20301->20312 20309->20305
## [26] 20315->20314 20315->20308 20319->20308 20319->20317 20319->20314
## [31] 20317->20304 20317->20307 20317->20318

# Subset vertices and edges
igraph::V(g0203)

## + 18/18 vertices, named, from c9a1236:
## [1] 20304 20314 20303 20305 20316 20320 20308 20313 20302 20312 20306 20318
## [13] 20307 20301 20309 20315 20319 20317

igraph::E(g0203)

## + 33/33 edges from c9a1236 (vertex names):
## [1] 20304->20318 20304->20317 20303->20301 20305->20313 20305->20301
## [6] 20316->20313 20316->20302 20316->20305 20320->20316 20320->20302
## [11] 20308->20304 20308->20317 20302->20306 20302->20305 20302->20314
## [16] 20312->20301 20312->20317 20306->20302 20318->20317 20318->20304
## [21] 20318->20319 20307->20304 20307->20319 20301->20312 20309->20305
## [26] 20315->20314 20315->20308 20319->20308 20319->20317 20319->20314
## [31] 20317->20304 20317->20307 20317->20318

```

```
# Count number of edges
igraph::ecount(g0203)

## [1] 33

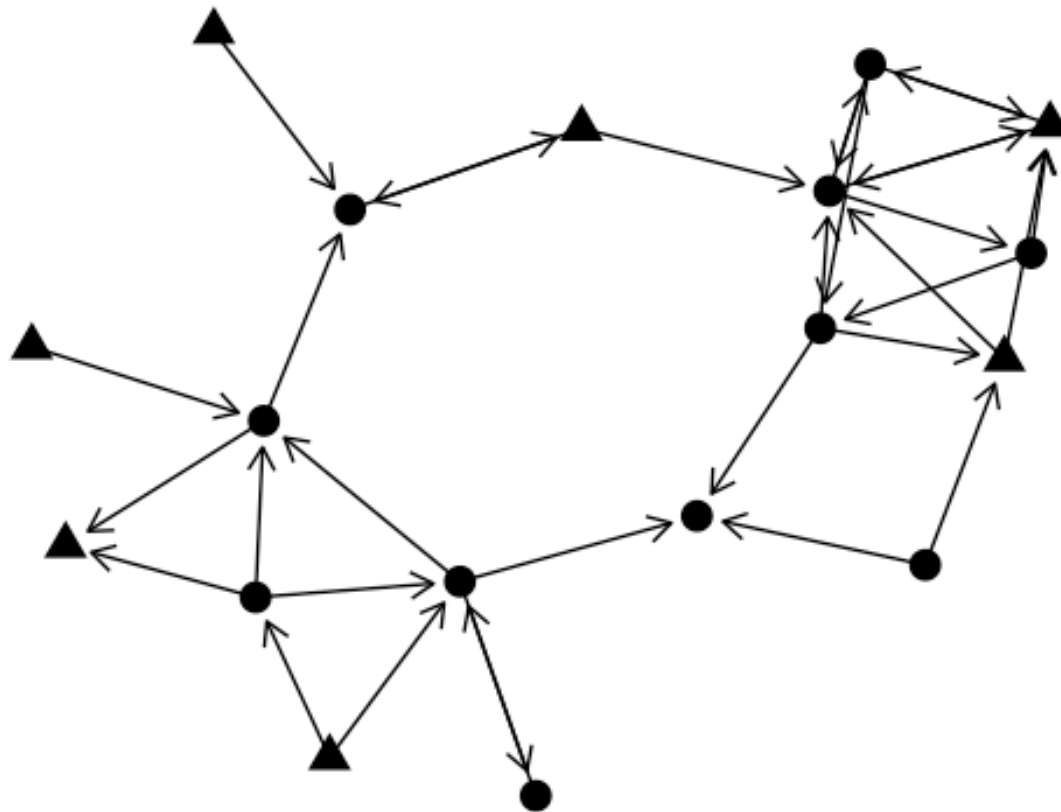
# Count number of vertices
igraph::vcount(g0203)

## [1] 18

#Assign risk status as a vertex attribute
igraph::V(g0203)$risk <- as.character(nodes0203$risk_status)

ggraph::ggraph(g0203, layout = "with_kk") +
  ggraph::geom_edge_link(arrow = ggplot2::arrow(length = ggplot2::unit(3, 'mm'), type = "open", ends =
"last"),
                        end_cap = ggraph::circle(3, 'mm')) +
  ggraph::geom_node_point(ggplot2::aes(shape = risk, size = 2)) + ggplot2::theme_void() +
ggplot2::theme(legend.position = "none") + ggplot2::ggtitle("Classroom 203")
```

Classroom 203



Classroom 0204

```
friends0204 <- dplyr::filter(langnet, stringr::str_detect(student_id, "^204"))

nodes0204 <- dplyr::select(friends0204, student_id, risk_status)

ties0204 <- dplyr::select(friends0204, nom = student_id, bf1, bf2, bf3) %>%
  tidyr::pivot_longer(-nom, values_to = "nom2") %>%
  dplyr::filter(., nom2 != "NA") %>%
```

```

dplyr::select(., -name)

g0204 <- igraph::graph_from_data_frame(ties0204, directed = TRUE, vertices = nodes0204)

## Warning in if (class(newval) == "factor") {: the condition has length > 1 and
## only the first element will be used

g0204

## IGRAPH 4fc3e40 DN-- 22 50 --
## + attr: name (v/c), risk_status (v/n)
## + edges from 4fc3e40 (vertex names):
## [1] 20414->20409 20414->20415 20405->20420 20405->20401 20416->20404
## [6] 20416->20410 20411->20401 20411->20416 20411->20412 20407->20420
## [11] 20407->20416 20420->20401 20420->20417 20420->20406 20401->20416
## [16] 20401->20404 20401->20420 20408->20421 20408->20404 20408->20410
## [21] 20406->20410 20406->20403 20413->20409 20413->20414 20413->20416
## [26] 20422->20406 20421->20410 20421->20412 20421->20416 20412->20401
## [31] 20412->20410 20412->20413 20410->20416 20410->20422 20419->20413
## [36] 20403->20412 20415->20418 20404->20416 20404->20410 20404->20412
## + ... omitted several edges

# Subset vertices and edges
igraph::V(g0204)

## + 22/22 vertices, named, from 4fc3e40:
## [1] 20414 20405 20416 20411 20407 20420 20401 20408 20406 20413 20422 20421
## [13] 20412 20410 20419 20403 20415 20404 20417 20409 20418 20402

igraph::E(g0204)

## + 50/50 edges from 4fc3e40 (vertex names):
## [1] 20414->20409 20414->20415 20405->20420 20405->20401 20416->20404
## [6] 20416->20410 20411->20401 20411->20416 20411->20412 20407->20420
## [11] 20407->20416 20420->20401 20420->20417 20420->20406 20401->20416
## [16] 20401->20404 20401->20420 20408->20421 20408->20404 20408->20410
## [21] 20406->20410 20406->20403 20413->20409 20413->20414 20413->20416

```

```
## [26] 20422->20406 20421->20410 20421->20412 20421->20416 20412->20401
## [31] 20412->20410 20412->20413 20410->20416 20410->20422 20419->20413
## [36] 20403->20412 20415->20418 20404->20416 20404->20410 20404->20412
## [41] 20417->20420 20417->20412 20417->20416 20409->20417 20409->20415
## [46] 20418->20405 20418->20415 20402->20415 20402->20418 20402->20406
```

Count number of edges

```
igraph::ecount(g0204)
```

```
## [1] 50
```

Count number of vertices

```
igraph::vcount(g0204)
```

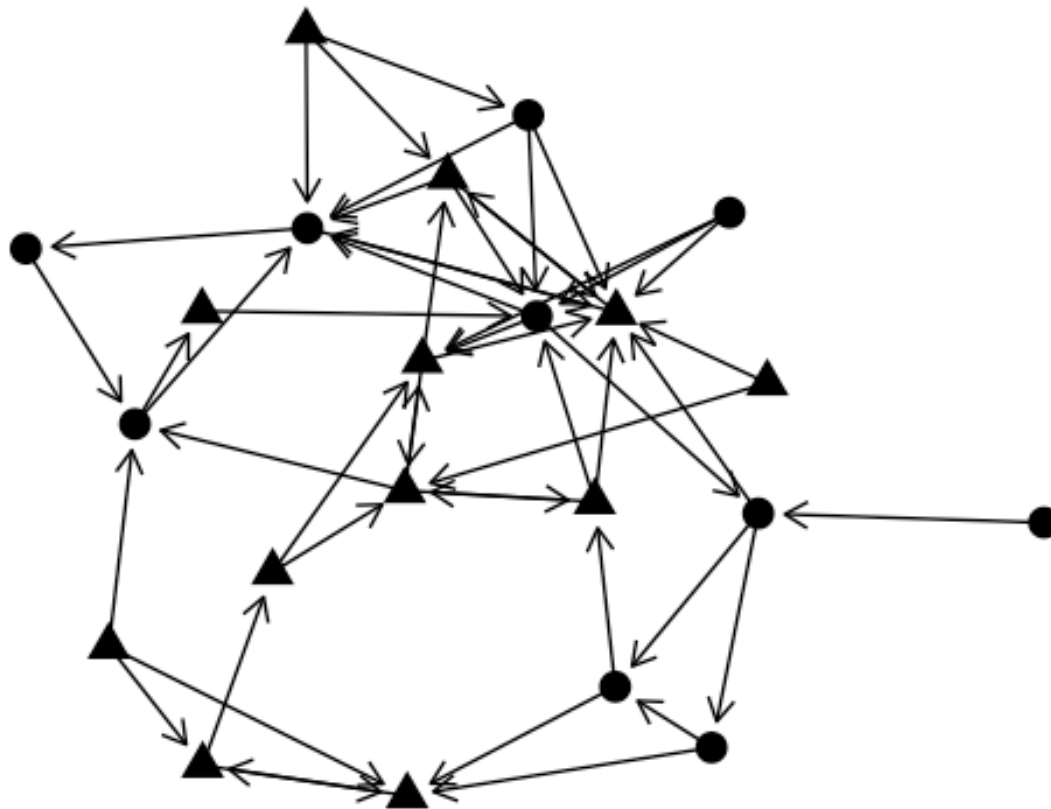
```
## [1] 22
```

#Assign risk status as a vertex attribute

```
igraph::V(g0204)$risk <- as.character(nodes0204$risk_status)
```

```
ggraph::ggraph(g0204, layout = "with_kk") +
  ggraph::geom_edge_link(arrow = ggplot2::arrow(length = ggplot2::unit(3, 'mm'), type = "open", ends =
    "last"),
    end_cap = ggraph::circle(3, 'mm')) +
  ggraph::geom_node_point(ggplot2::aes(shape = risk, size = 2)) + ggplot2::theme_void() +
  ggplot2::theme(legend.position = "none") + ggplot2::ggtitle("Classroom 204")
```

Classroom 204



Classroom 0401

```
friends0401 <- dplyr::filter(langnet, stringr::str_detect(student_id, "^401"))  
  
nodes0401 <- dplyr::select(friends0401, student_id, risk_status)  
  
ties0401 <- dplyr::select(friends0401, nom = student_id, bf1, bf2, bf3) %>%  
  tidyr::pivot_longer(-nom, values_to = "nom2") %>%  
  dplyr::filter(., nom2 != "NA") %>%
```

```

dplyr::select(., -name)

g0401 <- igraph::graph_from_data_frame(ties0401, directed = TRUE, vertices = nodes0401)

## Warning in if (class(newval) == "factor") {: the condition has length > 1 and
## only the first element will be used

g0401

## IGRAPH 9ae215c DN-- 21 49 --
## + attr: name (v/c), risk_status (v/n)
## + edges from 9ae215c (vertex names):
## [1] 40119->40113 40119->40120 40119->40105 40105->40106 40105->40107
## [6] 40105->40108 40118->40120 40118->40113 40118->40106 40107->40105
## [11] 40107->40106 40107->40109 40106->40116 40106->40108 40106->40105
## [16] 40109->40103 40109->40107 40109->40116 40120->40119 40102->40107
## [21] 40102->40109 40115->40101 40115->40113 40115->40120 40116->40105
## [26] 40116->40106 40116->40109 40113->40101 40113->40120 40101->40115
## [31] 40101->40121 40101->40109 40110->40109 40110->40106 40108->40106
## [36] 40108->40113 40108->40101 40103->40109 40112->40115 40112->40118
## + ... omitted several edges

# Subset vertices and edges
igraph::V(g0401)

## + 21/21 vertices, named, from 9ae215c:
## [1] 40119 40105 40118 40111 40107 40106 40109 40120 40102 40115 40116 40113
## [13] 40101 40110 40108 40103 40112 40121 40104 40114 40117

igraph::E(g0401)

## + 49/49 edges from 9ae215c (vertex names):
## [1] 40119->40113 40119->40120 40119->40105 40105->40106 40105->40107
## [6] 40105->40108 40118->40120 40118->40113 40118->40106 40107->40105
## [11] 40107->40106 40107->40109 40106->40116 40106->40108 40106->40105
## [16] 40109->40103 40109->40107 40109->40116 40120->40119 40102->40107
## [21] 40102->40109 40115->40101 40115->40113 40115->40120 40116->40105

```

```

## [26] 40116->40106 40116->40109 40113->40101 40113->40120 40101->40115
## [31] 40101->40121 40101->40109 40110->40109 40110->40106 40108->40106
## [36] 40108->40113 40108->40101 40103->40109 40112->40115 40112->40118
## [41] 40112->40119 40121->40120 40121->40103 40121->40107 40104->40110
## [46] 40104->40116 40104->40109 40114->40113 40117->40118

# Count number of edges
igraph::ecount(g0401)

## [1] 49

# Count number of vertices
igraph::vcount(g0401)

## [1] 21

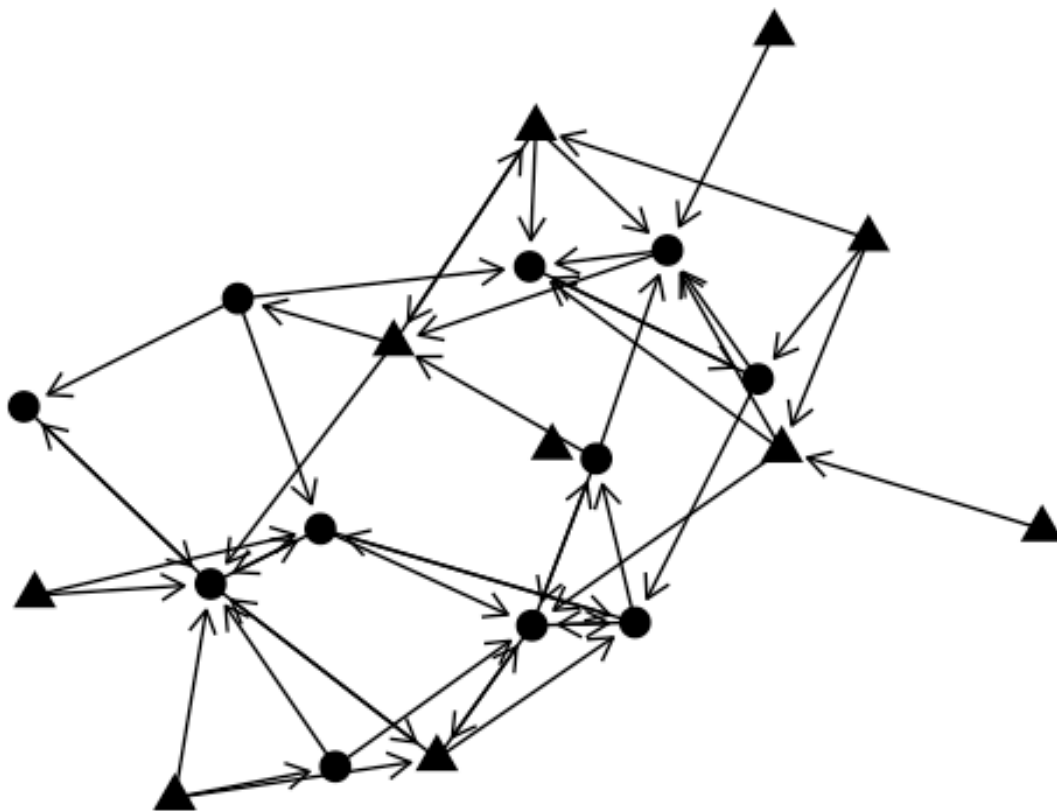
#Assign risk status as a vertex attribute
igraph::V(g0401)$risk <- as.character(nodes0401$risk_status)

class401 <- ggraph::ggraph(g0401, layout = "with_kk") +
  ggraph::geom_edge_link(arrow = ggplot2::arrow(length = ggplot2::unit(3, 'mm'), type = "open", ends =
"last"),
                        end_cap = ggraph::circle(3, 'mm')) +
  ggraph::geom_node_point(ggplot2::aes(shape = risk, size = 2)) + ggplot2::theme_void() +
ggplot2::theme(legend.position = "none") + ggplot2::ggtitle("Classroom 401")

class401

```

Classroom 401



Classroom 0403

```
friends0403 <- dplyr::filter(langnet, stringr::str_detect(student_id, "^403"))  
  
nodes0403 <- dplyr::select(friends0403, student_id, risk_status)  
  
ties0403 <- dplyr::select(friends0403, nom = student_id, bf1, bf2, bf3) %>%  
  tidyr::pivot_longer(-nom, values_to = "nom2") %>%  
  dplyr::filter(., nom2 != "NA") %>%
```

```

dplyr::select(., -name)

g0403 <- igraph::graph_from_data_frame(ties0403, directed = TRUE, vertices = nodes0403)

## Warning in if (class(newval) == "factor") {: the condition has length > 1 and
## only the first element will be used

g0403

## IGRAPH 4068367 DN-- 21 53 --
## + attr: name (v/c), risk_status (v/n)
## + edges from 4068367 (vertex names):
## [1] 40316->40310 40316->40312 40318->40312 40318->40301 40318->40321
## [6] 40309->40315 40309->40307 40309->40313 40303->40320 40303->40311
## [11] 40303->40313 40315->40307 40315->40309 40321->40312 40321->40316
## [16] 40310->40308 40310->40316 40310->40309 40319->40317 40319->40312
## [21] 40319->40318 40317->40310 40317->40303 40317->40319 40314->40321
## [26] 40314->40319 40301->40319 40305->40307 40305->40303 40305->40320
## [31] 40308->40311 40308->40308 40308->40306 40306->40317 40306->40308
## [36] 40306->40311 40312->40318 40312->40314 40312->40311 40311->40307
## + ... omitted several edges

# Subset vertices and edges
igraph::V(g0403)

## + 21/21 vertices, named, from 4068367:
## [1] 40316 40318 40309 40303 40315 40321 40310 40319 40317 40314 40301 40305
## [13] 40308 40306 40312 40311 40304 40313 40302 40307 40320

igraph::E(g0403)

## + 53/53 edges from 4068367 (vertex names):
## [1] 40316->40310 40316->40312 40318->40312 40318->40301 40318->40321
## [6] 40309->40315 40309->40307 40309->40313 40303->40320 40303->40311
## [11] 40303->40313 40315->40307 40315->40309 40321->40312 40321->40316
## [16] 40310->40308 40310->40316 40310->40309 40319->40317 40319->40312
## [21] 40319->40318 40317->40310 40317->40303 40317->40319 40314->40321

```

```

## [26] 40314->40319 40301->40319 40305->40307 40305->40303 40305->40320
## [31] 40308->40311 40308->40308 40308->40306 40306->40317 40306->40308
## [36] 40306->40311 40312->40318 40312->40314 40312->40311 40311->40307
## [41] 40311->40308 40304->40307 40304->40321 40304->40310 40313->40306
## [46] 40313->40308 40302->40320 40302->40311 40302->40313 40307->40315
## + ... omitted several edges

# Count number of edges
igraph::ecount(g0403)

## [1] 53

# Count number of vertices
igraph::vcount(g0403)

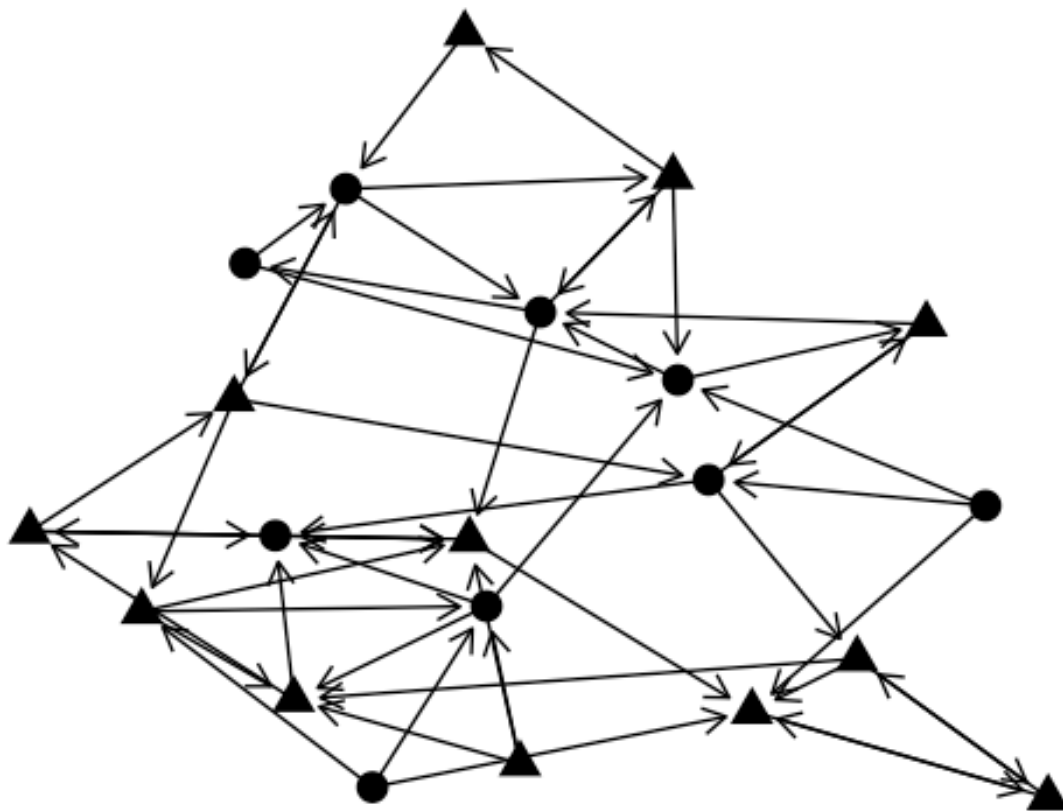
## [1] 21

#Assign risk status as a vertex attribute
igraph::V(g0403)$risk <- as.character(nodes0403$risk_status)

ggraph::ggraph(g0403, layout = "with_kk") +
  ggraph::geom_edge_link(arrow = ggplot2::arrow(length = ggplot2::unit(3, 'mm'), type = "open", ends =
"last"),
                        end_cap = ggraph::circle(3, 'mm')) +
  ggraph::geom_node_point(ggplot2::aes(shape = risk, size = 2)) + ggplot2::theme_void() +
ggplot2::theme(legend.position = "none") + ggplot2::ggtitle("Classroom 403")

```

Classroom 403



Create a Combined Plot of 4 Classrooms...

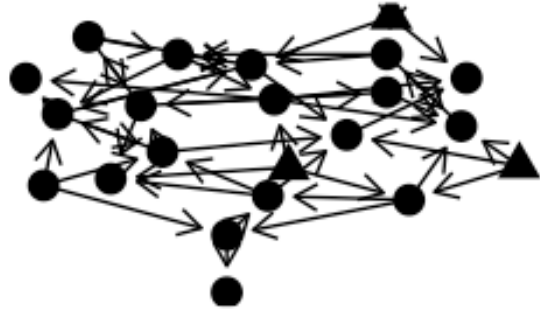
```
p <- (class101|class103)/  
(class201|class401) + plot_annotation(  
  title = 'Classroom Friendship Networks',  
  subtitle = 'by Student Risk Status',  
  caption = 'Students at risk of language disorder are denoted with triangles. Student NOT at risk are  
denoted with a circles. Arrows indicate the direction of each friendship nomination.')

```

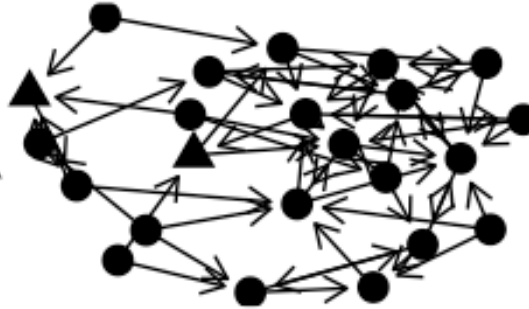
p

Classroom Friendship Networks by Student Risk Status

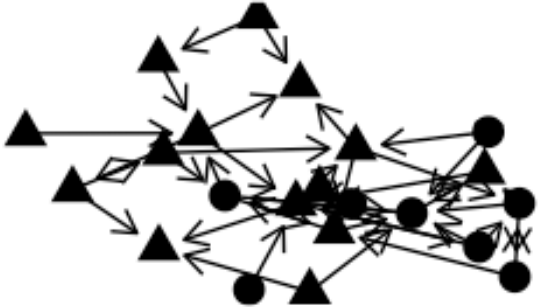
Classroom 101



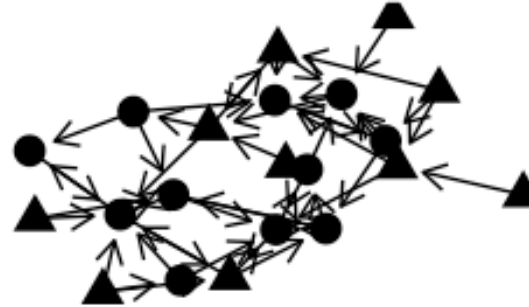
Classroom 103



Classroom 201



Classroom 401



: risk are denoted with a circles. Arrows indicate the direction of each friendship nomination.

Calculating Network-Level Descriptives for Each Classroom

```
classes <- list(g0101, g0103, g0105, g0106, g0201, g0203, g0204, g0401, g0403)
```

```
size <- classes %>%  
  purrr::map(igraph::vcount)
```

```

n_edges <- classes %>%
  purrr::map(igraph::ecount)

density <- classes %>%
  purrr::map(igraph::edge_density)

recip <- classes %>%
  purrr::map(igraph::reciprocity)

distance = classes %>%
  purrr::map(igraph::mean_distance)

net_summary <- cbind(c("Class 101", "Class 103", "Class 105", "Class 106", "Class 201", "Class 203", "Class
204", "Class 401", "Class 403"),
  as.matrix(unlist(size), ncol = 1),
  as.matrix(unlist(n_edges), ncol = 1),
  as.matrix(unlist(density), ncol = 1),
  as.matrix(unlist(recip), ncol = 1),
  as.matrix(unlist(distance), ncol = 1)) %>%
  dplyr::as_tibble(.name_repair = ~ c("class", "size", "edges", "density", "recip", "distance"))

```

Calculating Degree Centrality Measures for Each Classroom Network

Important Note: Nodes without ties are left in here. To get the centrality to calculate, I adjusted the alpha from 1 to .999.

```

cent_0101 <- igraph::degree(g0101, normalized = TRUE)
central0101 <- dplyr::tibble(student_id = names(cent_0101), cent = cent_0101)

cent_0103 <- igraph::degree(g0103, normalized = TRUE)
central0103 <- dplyr::tibble(student_id = names(cent_0103), cent = cent_0103)

cent_0105 <- igraph::degree(g0105, normalized = TRUE)
central0105 <- dplyr::tibble(student_id = names(cent_0105), cent = cent_0105)

```

```

cent_0106 <- igraph::degree(g0106, normalized = TRUE)
central0106 <- dplyr::tibble(student_id = names(cent_0106), cent = cent_0106)

cent_0201 <- igraph::degree(g0201, normalized = TRUE)
central0201 <- dplyr::tibble(student_id = names(cent_0201), cent = cent_0201)

cent_0203 <- igraph::degree(g0203, normalized = TRUE)
central0203 <- dplyr::tibble(student_id = names(cent_0203), cent = cent_0203)

cent_0204 <- igraph::degree(g0204, normalized = TRUE)
central0204 <- dplyr::tibble(student_id = names(cent_0204), cent = cent_0204)

cent_0401 <- igraph::degree(g0401, normalized = TRUE)
central0401 <- dplyr::tibble(student_id = names(cent_0401), cent = cent_0401)

cent_0403 <- igraph::degree(g0403, normalized = TRUE)
central0403 <- dplyr::tibble(student_id = names(cent_0403), cent = cent_0403)

cent <- dplyr::bind_rows(central0101, central0103, central0105, central0106,
                        central0201, central0203, central0204, central0401, central0403)

```

Calculating Closeness Centrality Measures for Each Classroom Network

```

cent_0101b <- igraph::closeness(g0101, normalized = TRUE)

## Warning in igraph::closeness(g0101, normalized = TRUE): At centrality.c:
## 2784 :closeness centrality is not well-defined for disconnected graphs

central0101b <- dplyr::tibble(student_id = names(cent_0101b), centb = cent_0101b)

cent_0103b <- igraph::closeness(g0103, normalized = TRUE)

## Warning in igraph::closeness(g0103, normalized = TRUE): At centrality.c:
## 2784 :closeness centrality is not well-defined for disconnected graphs

```

```
central0103b <- dplyr::tibble(student_id = names(cent_0103b), centb = cent_0103b)

cent_0105b <- igraph::closeness(g0105, normalized = TRUE)
## Warning in igraph::closeness(g0105, normalized = TRUE): At centrality.c:
## 2784 :closeness centrality is not well-defined for disconnected graphs

central0105b <- dplyr::tibble(student_id = names(cent_0105b), centb = cent_0105b)

cent_0106b <- igraph::closeness(g0106, normalized = TRUE)
## Warning in igraph::closeness(g0106, normalized = TRUE): At centrality.c:
## 2784 :closeness centrality is not well-defined for disconnected graphs

central0106b <- dplyr::tibble(student_id = names(cent_0106b), centb = cent_0106b)

cent_0201b <- igraph::closeness(g0201, normalized = TRUE)
## Warning in igraph::closeness(g0201, normalized = TRUE): At centrality.c:
## 2784 :closeness centrality is not well-defined for disconnected graphs

central0201b <- dplyr::tibble(student_id = names(cent_0201b), centb = cent_0201b)

cent_0203b <- igraph::closeness(g0203, normalized = TRUE)
## Warning in igraph::closeness(g0203, normalized = TRUE): At centrality.c:
## 2784 :closeness centrality is not well-defined for disconnected graphs

central0203b <- dplyr::tibble(student_id = names(cent_0203b), centb = cent_0203b)

cent_0204b <- igraph::closeness(g0204, normalized = TRUE)
## Warning in igraph::closeness(g0204, normalized = TRUE): At centrality.c:
## 2784 :closeness centrality is not well-defined for disconnected graphs

central0204b <- dplyr::tibble(student_id = names(cent_0204b), centb = cent_0204b)

cent_0401b <- igraph::closeness(g0401, normalized = TRUE)
```

```
## Warning in igraph::closeness(g0401, normalized = TRUE): At centrality.c:
## 2784 :closeness centrality is not well-defined for disconnected graphs

central0401b <- dplyr::tibble(student_id = names(cent_0401b), centb = cent_0401b)

cent_0403b <- igraph::closeness(g0403, normalized = TRUE)

## Warning in igraph::closeness(g0403, normalized = TRUE): At centrality.c:
## 2784 :closeness centrality is not well-defined for disconnected graphs

central0403b <- dplyr::tibble(student_id = names(cent_0403b), centb = cent_0403b)

centb <- dplyr::bind_rows(central0101b, central0103b, central0105b, central0106b,
                          central0201b, central0203b, central0204b, central0401b, central0403b)
```

Merge centrality scores with the rest of the dataset

```
net_final <- dplyr::full_join(cent, langnet, by = "student_id") %>%
  dplyr::full_join(., centb, by = "student_id")

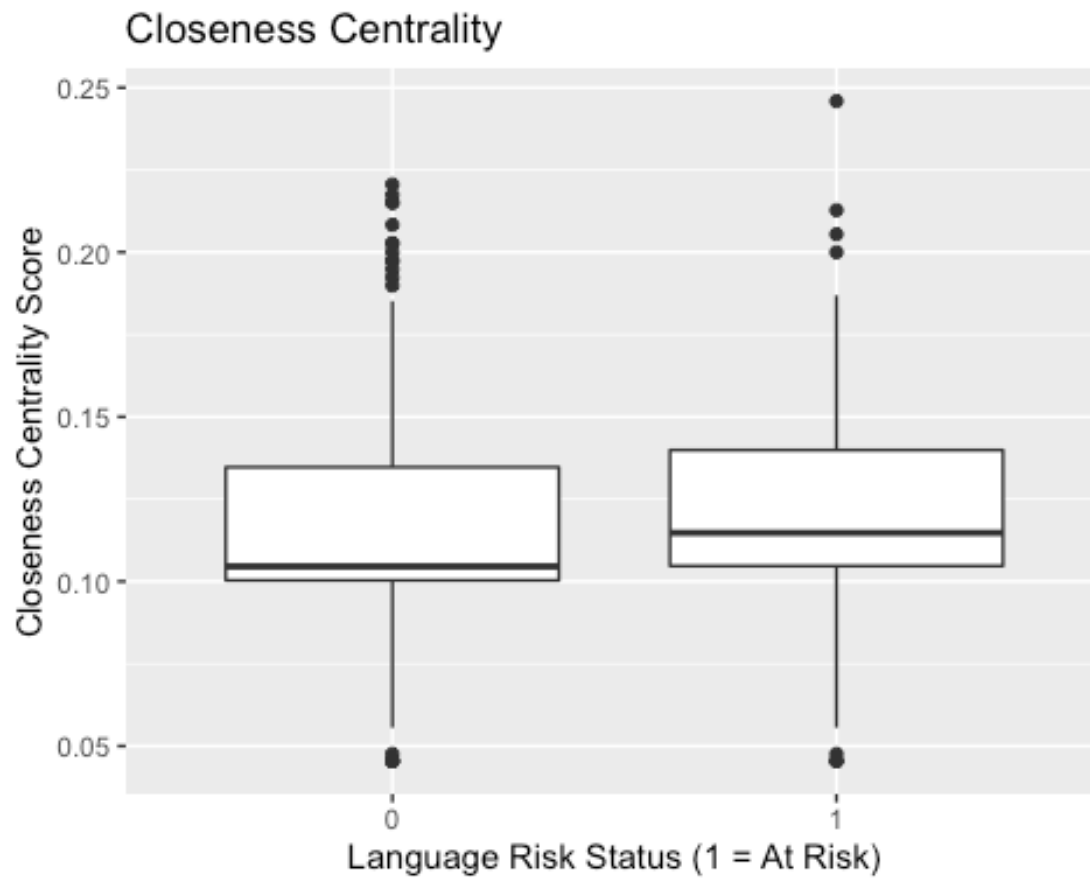
dplyr::glimpse(net_final)

## Rows: 188
## Columns: 16
## $ student_id      <chr> "10108", "10117", "10119", "10102", "10114", "10107", ...
## $ cent             <dbl> 0.28571429, 0.28571429, 0.14285714, 0.19047619, 0.1428...
## $ teachid         <dbl+lbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
## $ bf1             <chr> "10120", "10101", "NA", "10117", "10106", "10116", "NA...
## $ bf2             <chr> "NA", "10102", "NA", "10108", "10103", "10104", "NA", ...
## $ bf3             <chr> "NA", "10119", "NA", "10120", "10119", "10105", "NA", ...
## $ risk_status     <dbl+lbl> 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, ...
## $ celf_score      <dbl> 14, 10, 12, 12, 2, 11, 8, 12, 12, 15, 4, 12, 7, 10, 15...
## $ race_eth        <dbl+lbl> 1, 1, 1, 1, 5, 1, 1, 1, 1, 1, 1, 1, 1, 5, 1, 1, ...
## $ gen             <dbl+lbl> 2, 1, 2, 1, 2, 1, 2, 2, 2, 1, 1, 2, 1, 2, 1, 2, ...
## $ csize           <dbl> 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22...
## $ age_yrs         <dbl> 6.592745, 6.516085, 6.425736, 7.071868, 6.710472, 6.34...
```

```
## $ risk_status_fac <fct> 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, ...
## $ race_eth_fac      <fct> WHITE, WHITE, WHITE, WHITE, OTHER, WHITE, WHITE, WHITE...
## $ gen_fac           <fct> MALE, FEMALE, MALE, FEMALE, MALE, FEMALE, MALE, MALE, ...
## $ centb             <dbl> 0.09333333, 0.10096154, 0.04545455, 0.10096154, 0.1794...
```

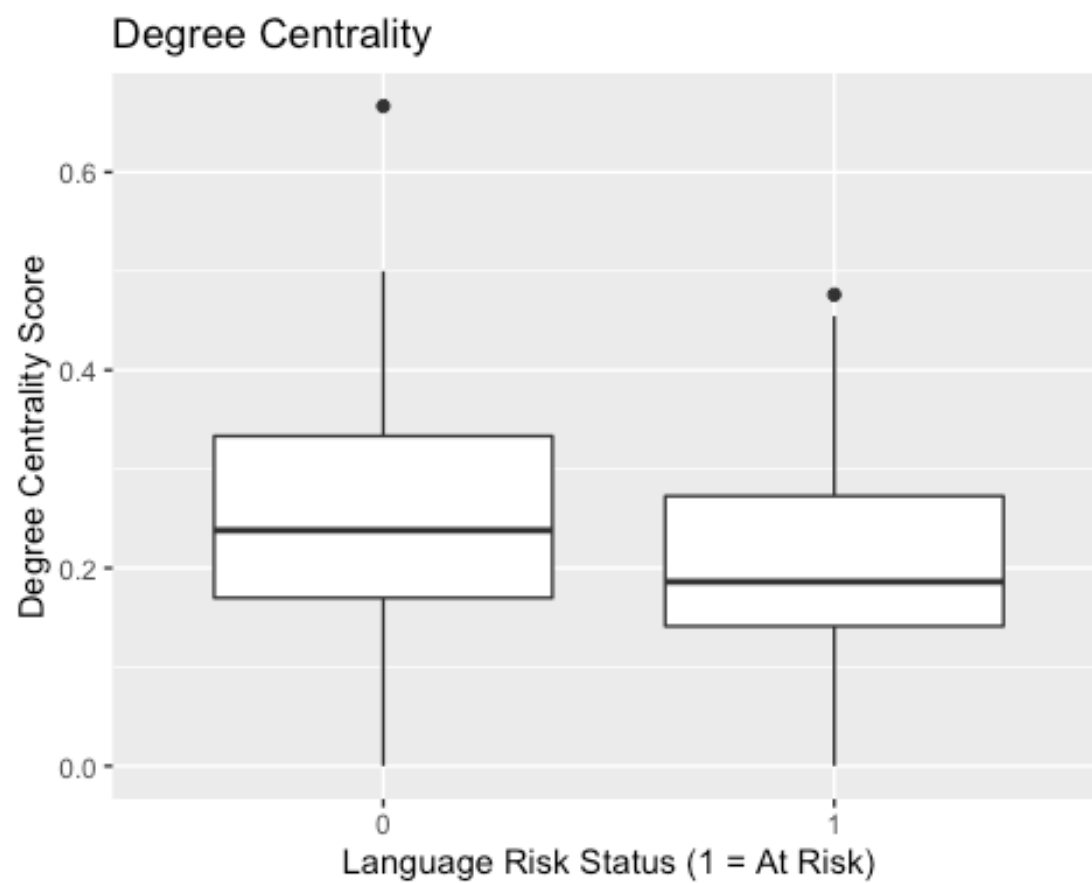
Create a Combined Plot Degree and Closeness Centrality Histograms...

```
close <- ggplot2::ggplot(net_final, ggplot2::aes(risk_status_fac, centb)) +
  ggplot2::geom_boxplot() + ggplot2::labs(x = "Language Risk Status (1 = At Risk)",
    y = "Closeness Centrality Score",
    title = "Closeness Centrality")
close
```



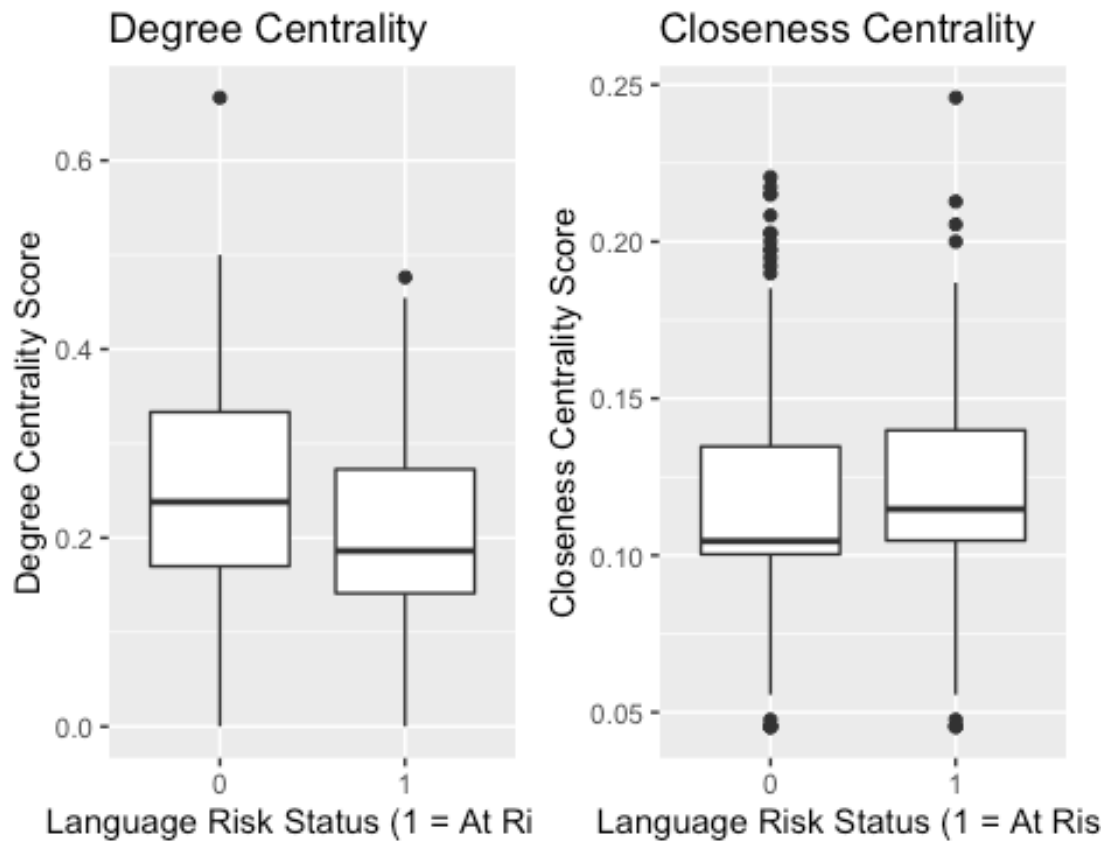
```
degree <- ggplot2::ggplot(net_final, ggplot2::aes(risk_status_fac, cent)) +  
  ggplot2::geom_boxplot() + ggplot2::labs(x = "Language Risk Status (1 = At Risk)",  
    y = "Degree Centrality Score",  
    title = "Degree Centrality")
```

degree



```
p2 <- degree + close
```

```
p2
```



Now, we can do some modeling! We have one complete dataset, with centrality and student_characteristics.

Regression Models for Degree Centrality

Binary IV of Interest (Risk_Status)

```
model.1 <- lm(cent ~ risk_status_fac + gen_fac + age_yrs + race_eth_fac, data = na.omit(net_final))
summary(model.1)
```

```
##
## Call:
## lm(formula = cent ~ risk_status_fac + gen_fac + age_yrs + race_eth_fac,
##     data = na.omit(net_final))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.23236 -0.07647 -0.01740  0.07544  0.35196
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.277840    0.174803  -1.589   0.11376
## risk_status_fac1 -0.050670    0.017511  -2.894   0.00429 **
## gen_facMALE    -0.025522    0.016553  -1.542   0.12492
## age_yrs        0.081853    0.025775   3.176   0.00177 **
## race_eth_facAFRICAN AMERICAN -0.030975    0.023440  -1.321   0.18808
## race_eth_facASIAN -0.028518    0.056715  -0.503   0.61572
## race_eth_facHISPANIC -0.060228    0.033944  -1.774   0.07774 .
## race_eth_facOTHER  0.001027    0.028130   0.037   0.97091
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1107 on 175 degrees of freedom
## Multiple R-squared:  0.1228, Adjusted R-squared:  0.08775
## F-statistic: 3.501 on 7 and 175 DF,  p-value: 0.001537
```

Continuous IV of Interest (CELF_Score)

```
model.2 <- lm(cent ~ celf_score + gen_fac + age_yrs + race_eth_fac, data = na.omit(net_final))
summary(model.2)

##
## Call:
## lm(formula = cent ~ celf_score + gen_fac + age_yrs + race_eth_fac,
##     data = na.omit(net_final))
##
## Residuals:
```

```
##      Min      1Q   Median      3Q      Max
## -0.23367 -0.08177 -0.01924  0.06840  0.35796
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -0.261207   0.174655  -1.496  0.13657
## celf_score      0.008052   0.002856   2.819  0.00537 **
## gen_facMALE    -0.026885   0.016546  -1.625  0.10601
## age_yrs        0.065173   0.025832   2.523  0.01253 *
## race_eth_facAFRICAN AMERICAN -0.030235   0.023487  -1.287  0.19969
## race_eth_facASIAN    -0.026886   0.056811  -0.473  0.63662
## race_eth_facHISPANIC -0.060570   0.034007  -1.781  0.07663 .
## race_eth_facOTHER     0.002995   0.028230   0.106  0.91564
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1108 on 175 degrees of freedom
## Multiple R-squared:  0.1208, Adjusted R-squared:  0.08563
## F-statistic: 3.435 on 7 and 175 DF,  p-value: 0.001813
```

Multilevel Models for Closeness Centrality

Binary IV of Interest (Risk_Status)

```
model.1b <- lm(cenrb ~ risk_status_fac + gen_fac + age_yrs + race_eth_fac, data = net_final)
summary(model.1b)

##
## Call:
## lm(formula = cenrb ~ risk_status_fac + gen_fac + age_yrs + race_eth_fac,
##     data = net_final)
##
## Residuals:
##      Min      1Q   Median      3Q      Max
## -0.097830 -0.021646 -0.009471  0.014008  0.103812
```

```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.0355900   0.0632295    0.563   0.5742
## risk_status_fac1 -0.0004097   0.0063340   -0.065   0.9485
## gen_facMALE      0.0061520   0.0059876    1.027   0.3056
## age_yrs          0.0114859   0.0093234    1.232   0.2196
## race_eth_facAFRICAN AMERICAN 0.0164066   0.0084787    1.935   0.0546 .
## race_eth_facASIAN  -0.0098665   0.0205150   -0.481   0.6312
## race_eth_facHISPANIC 0.0197196   0.0122780    1.606   0.1101
## race_eth_facOTHER    0.0142906   0.0101750    1.404   0.1619
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.04005 on 175 degrees of freedom
## (5 observations deleted due to missingness)
## Multiple R-squared:  0.05619,    Adjusted R-squared:  0.01844
## F-statistic: 1.488 on 7 and 175 DF,  p-value: 0.1741
```

Continuous IV of Interest (CELF_Score)

```
model.2b <- lm(cenb ~ celf_score + gen_fac + age_yrs + race_eth_fac, data = net_final)
summary(model.2b)
```

```
##
## Call:
## lm(formula = cenb ~ celf_score + gen_fac + age_yrs + race_eth_fac,
##     data = net_final)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.097408 -0.021439 -0.009184  0.013612  0.103046
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.0377313   0.0630369    0.599   0.5502
## celf_score     -0.0006255   0.0010309   -0.607   0.5448
```

```
## gen_facMALE          0.0059665  0.0059720  0.999  0.3191
## age_yrs              0.0120864  0.0093235  1.296  0.1966
## race_eth_facAFRICAN AMERICAN 0.0159958  0.0084772  1.887  0.0608 .
## race_eth_facASIAN     -0.0105613  0.0205043 -0.515  0.6071
## race_eth_facHISPANIC   0.0176361  0.0122738  1.437  0.1525
## race_eth_facOTHER      0.0135998  0.0101890  1.335  0.1837
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.04001 on 175 degrees of freedom
## (5 observations deleted due to missingness)
## Multiple R-squared:  0.05815,    Adjusted R-squared:  0.02048
## F-statistic: 1.544 on 7 and 175 DF,  p-value: 0.1555
```

Using Bootstrapping to Remove Dependency in Positional Network Measures

Binary IV of Interest (Risk_Status)

```
set.seed(23284) # for reproducibility
```

```
boot.model.1 <- car::Boot(model.1, R = 1000)
```

```
## Loading required namespace: boot
```

```
summary(boot.model.1)
```

```
##
## Number of bootstrap replications R = 980
##
##              original    bootBias    bootSE    bootMed
## (Intercept)   -0.2778404  4.5771e-03  0.210343 -2.7305e-01
## risk_status_fac1 -0.0506697  4.5828e-04  0.017304 -4.9726e-02
## gen_facMALE    -0.0255222  8.9022e-04  0.016721 -2.4158e-02
## age_yrs        0.0818533 -7.4152e-04  0.031044  8.1451e-02
## race_eth_facAFRICAN AMERICAN -0.0309751  1.1022e-03  0.022564 -3.0172e-02
## race_eth_facASIAN -0.0285183 -5.7404e-04  0.0098116 -3.0891e-02
## race_eth_facHISPANIC -0.0602284 -5.7897e-05  0.027367 -5.9936e-02
## race_eth_facOTHER 0.0010273 -1.1484e-03  0.024683  1.5144e-06
```

```
car::Confint(boot.model.1, level = .95, type = "perc")

## Bootstrap percent confidence intervals
##
##               Estimate      2.5 %      97.5 %
## (Intercept)      -0.277840390 -0.67865282  0.137931579
## risk_status_fac1  -0.050669699 -0.08531562 -0.017414790
## gen_facMALE       -0.025522186 -0.06029287  0.008131123
## age_yrs           0.081853286  0.01962168  0.141154471
## race_eth_facAFRICAN AMERICAN -0.030975127 -0.07371534  0.015561385
## race_eth_facASIAN  -0.028518256 -0.20428186  0.203515619
## race_eth_facHISPANIC -0.060228438 -0.11320604 -0.003229815
## race_eth_facOTHER  0.001027284 -0.04672167  0.049966486
```

Continuous IV of Interest (CELF_Score)

```
set.seed(23284) # for reproducibility
boot.model.2 <- car::Boot(model.2, R = 1000)
summary(boot.model.2)

##
## Number of bootstrap replications R = 980
##               original    bootBias    bootSE    bootMed
## (Intercept)      -0.2612069  5.2223e-03 0.2099832 -0.2584754
## celf_score        0.0080524 -4.3345e-05 0.0026891  0.0080930
## gen_facMALE       -0.0268845  8.5654e-04 0.0165869 -0.0254456
## age_yrs           0.0651726 -7.4548e-04 0.0307096  0.0638615
## race_eth_facAFRICAN AMERICAN -0.0302354  1.0182e-03 0.0228377 -0.0290137
## race_eth_facASIAN  -0.0268856 -1.9300e-04 0.0990629 -0.0265302
## race_eth_facHISPANIC -0.0605701 -9.2724e-05 0.0270398 -0.0604770
## race_eth_facOTHER  0.0029945 -7.3829e-04 0.0247342  0.0022273

car::Confint(boot.model.2, level = .95, type = "perc")

## Bootstrap percent confidence intervals
##
##               Estimate      2.5 %      97.5 %
```

## (Intercept)	-0.261206876	-0.671587480	0.159165130
## celf_score	0.008052394	0.002987236	0.013086319
## gen_facMALE	-0.026884513	-0.060111079	0.007440595
## age_yrs	0.065172559	0.004458352	0.124261742
## race_eth_facAFRICAN AMERICAN	-0.030235388	-0.074280170	0.017414046
## race_eth_facASIAN	-0.026885641	-0.205473210	0.215880882
## race_eth_facHISPANIC	-0.060570096	-0.114349118	-0.006598907
## race_eth_facOTHER	0.002994548	-0.046684777	0.052369912